PYTHON FOUNDATIONS - ASSIGNMENT 04

Logical Operators & Decision Making

15 scenario-based Python challenges combining **and**, **or** and **not** with comparison operators, if, elif and else.

and | or | not | if | elif | else | True / False | Real-world decisions

Name: _____

Date: _____

Batch: _____

BEGINNER TRACK
15 MULTI-PART CHALLENGES

Quick reference before you begin

Logical operators combine or reverse conditions. First create conditions using comparison operators such as `>=`, `==` or `!=`. Then connect those conditions using `and`, `or` or `not`.

and

True only when BOTH conditions are True

or

True when AT LEAST ONE condition is True

not

Reverses True to False and False to True

A	B	A and B	A or B
True	True	True	True
True	False	False	True
False	True	False	True
False	False	False	False

1 **Use `and` for mandatory requirements**

Example: marks >= 40 `and` attendance >= 75.

2 **Use `or` for alternatives**

Example: coupon is valid `or` customer is premium.

3 **Use `not` to reverse a Boolean**

Example: `not is_blocked` means the user is not blocked.

4 **Use parentheses when combining logic**

Example: `(age >= 18 and has_id) or has_guardian`.

Important: Do not use loops, lists, dictionaries or functions for this assignment. Focus only on input, type conversion, arithmetic, comparison, logical operators and conditional statements.

Both conditions must be true

Use `and` when every listed requirement is compulsory.

1 Student Exam Eligibility

AND | IF-ELSE | AVERAGE

A school allows a student to pass only when academic performance and attendance are both acceptable.

1. Take marks in Maths, Science and English and calculate the average.
2. Take attendance percentage.
3. If average is at least 40 **and** attendance is at least 75, print *PASS - Eligible*.
4. Otherwise print *Not Eligible*.
5. Also print total marks, rounded average and attendance.
6. Add an `elif` : if average is at least 40 but attendance is below 75, print *Academically passed, attendance shortage*.

2 Loan Pre-Eligibility Checker

AND | IF-ELIF-ELSE

A bank performs a simple preliminary eligibility check before processing a loan application.

1. Take applicant age, monthly salary and credit score.
2. Approve the preliminary check only if age is at least 21, salary is at least 30,000 and credit score is at least 700.
3. If age and salary qualify but credit score is below 700, print *Credit score needs improvement*.
4. Otherwise print *Not eligible for preliminary check*.
5. Print all three entered values in the final summary.

3 Secure Office Login

AND == IF-ELSE

An employee can enter a secure dashboard only when both username and PIN are correct.

1. Store a correct username and correct PIN in variables.
2. Take username and PIN from the user.
3. If both match, print *Login successful*.
4. Otherwise print *Invalid credentials*.
5. Add a second condition requiring `account_active == True`.
6. If credentials are correct but the account is inactive, print a separate message using `elif`.

4 Theme Park Ride Safety Check

AND RANGE IF-ELSE

A ride is allowed only when the visitor meets both age and height requirements.

1. Take visitor age and height in centimetres.
2. Allow the ride only when age is at least 12 **and** height is at least 140 cm.
3. If age qualifies but height does not, print *Height requirement not met*.
4. If height qualifies but age does not, print *Age requirement not met*.
5. Otherwise print *Ride allowed* only for the fully valid case.

One valid condition can be enough

Use `or` when satisfying any one of the allowed choices should produce a True result.

5 Online Store Discount Access

OR | IF-ELSE | DISCOUNT

A shopper receives a 10% discount if the order is large enough or if they are a premium member.

1. Take order amount.
2. Take premium-member status as `yes` or `no`.
3. If order amount is at least 5000 **or** the customer is premium, apply a 10% discount.
4. Otherwise keep the original bill.
5. Print original bill, discount amount and final bill.
6. Add an `elif`: if order amount is at least 10,000 and the member is premium, give 15% instead.

6 Emergency Entry Permission

OR | == | IF-ELSE

A restricted facility normally requires an access card, but emergency staff are also allowed entry.

1. Take whether the person has a valid access card.
2. Take the person's role: employee, doctor, firefighter or visitor.
3. Allow entry if the access card is valid **or** role is doctor **or** firefighter.
4. Otherwise print *Entry denied*.
5. Print the exact reason for the granted access in a separate message.

7 Free Delivery Rule

OR AND IF-ELIF-ELSE

An e-commerce app decides whether delivery is free, discounted or charged normally.

1. Take cart value and membership status.
2. Give free delivery if cart value is at least 999 **or** the user is a premium member.
3. If cart value is between 499 and 998 and the user is not premium, charge a reduced delivery fee of 40.
4. Otherwise charge a normal delivery fee of 80.
5. Print cart value, delivery fee and final payable amount.

8 Sports Team Selection

OR AND IF-ELIF

A student can qualify for the sports team through performance in either speed or endurance, provided attendance is acceptable.

1. Take 100-metre race time, endurance score and attendance percentage.
2. Qualify if race time is 14 seconds or less **or** endurance score is at least 80.
3. Additionally, attendance must be at least 75.
4. If sporting criteria pass but attendance fails, print *Performance qualified - attendance insufficient*.
5. Otherwise print selected/not selected appropriately.

Reverse a Boolean decision

Use `not` when your decision depends on something being false, absent or disabled.

9 Blocked Account Access

NOT AND BOOLEAN

A user may access an application only if the password is correct and the account is not blocked.

1. Store the correct password in a variable.
2. Take the user's password.
3. Take blocked status as `True` or `False` in the program.
4. Allow access when the password matches **and** `not is_blocked`.
5. If password is correct but the account is blocked, print *Contact support*.
6. Otherwise print *Incorrect password*.

10 Library Book Issue Check

NOT AND IF-ELSE

A library issues a book only when the member is active and does not have an overdue book.

1. Take member-active status and overdue-book status.
2. Issue the book if member is active **and** the user does **not** have an overdue book.
3. If membership is inactive, print *Renew membership first*.
4. If membership is active but overdue exists, print *Return overdue book first*.
5. Use `not` explicitly in at least one condition.

Combine AND, OR and NOT

Use parentheses to make the intended grouping of conditions easy to read.

11 Movie Ticket Age Rule

AND | OR | PARENTHESES

A cinema allows entry to an age-restricted movie under two possible paths.

1. Take age, whether the person has valid ID, and whether a guardian is present.
2. Allow normal entry when age is at least 18 **and** valid ID is available.
3. Also allow a 16-17 year old when a guardian is present.
4. Use parentheses to combine the two valid paths clearly.
5. Print *Entry allowed* or *Entry denied*.
6. Add a special message when age is below 16.

12 Employee Work-from-Home Eligibility

AND | OR | NOT | IF-ELIF

A company allows work from home based on role, performance and security restrictions.

1. Take employee role, performance rating and whether the employee is under a security restriction.
2. Allow WFH when role is developer **or** analyst, performance rating is at least 3, and the employee is **not** restricted.
3. If role and performance qualify but restriction is active, print *WFH blocked by security restriction*.
4. Otherwise print *WFH not eligible*.
5. Use parentheses around the role alternatives.

13 Hospital Appointment Priority

AND OR IF-ELIF-ELSE

A clinic assigns priority based on age, temperature and emergency status.

1. Take patient age, temperature and emergency status.
2. Assign **High Priority** if emergency is yes **or** temperature is 103 or above.
3. Assign **Priority** if age is 65 or above **and** temperature is at least 100.
4. Otherwise assign **Normal Queue**.
5. Print all patient details and the assigned category.
6. Ensure the high-priority rule is checked before the lower-priority rule.

14 Scholarship Decision System

AND OR ELIF PERCENTAGE

A scholarship program evaluates academic performance, family income and sports achievement.

1. Take three subject marks and calculate percentage.
2. Take annual family income and sports-achievement status.
3. Give **Full Scholarship** if percentage is at least 90 **and** family income is at most 500,000.
4. Give **Merit Scholarship** if percentage is at least 85 **or** sports achievement is yes.
5. Otherwise print *Not eligible for scholarship*.
6. Print percentage rounded to 2 decimal places along with the final decision.

15 Airport Boarding Eligibility System

AND OR NOT IF-ELIF-ELSE MASTER

Build a simplified airport boarding checker that combines identity, ticket, baggage and special-assistance rules.

1. Take passenger name, age, ticket status, valid-ID status, baggage weight and special-assistance status.
2. A normal adult passenger may board only when ticket is valid **and** ID is valid **and** baggage is at most 20 kg.
3. A passenger below 18 may board when ticket is valid **and** a guardian is present.
4. If baggage exceeds 20 kg but all other normal-adult requirements pass, print *Pay excess baggage fee before boarding*.
5. If ticket is not valid, deny boarding regardless of other values.
6. If special assistance is requested and boarding is otherwise allowed, print an additional assistance message.
7. Use at least one `and`, one `or` and one `not` somewhere in your final program.
8. Print a final boarding summary containing passenger name, baggage weight and decision.

Think before coding: Which rule should be checked first so that an invalid ticket can never accidentally reach an allowed branch?

Completion checklist

- I can explain the difference between `and` and `or`.
- I can explain what `not` does.
- I can combine comparisons with logical operators.
- I can use logical expressions inside `if`.
- I can choose between `if`, `elif` and `else`.
- I can use parentheses to make complex logic readable.
- I can predict whether a condition becomes True or False.
- I can solve a scenario without using loops or functions.

Teacher checkpoint: A student is ready for the next topic when they can translate a sentence such as "age is at least 18 and ID is valid" into a correct Python Boolean expression without help.